

GENERAL SPECIFICATION G8.3

Doors and Windows

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G8.3 DOORS AND WINDOWS

G8.3.1 Overview

This section covers the requirements for external and internal windows and doors including fire doors and their associated ironmongery.

G8.3.2 Applicable Standards

- (a) Unless otherwise agreed with the S.O., all doors and accessories shall comply with the relevant requirements of the Standards and Codes listed below or referenced in the body of the Specification. Alternative Standards and Codes may be submitted to the S.O., provided it can be demonstrated that the alternative Standards and Codes comply with the requirements of the standards specified. All Standards and Codes quoted are the current version, unless specific year references are noted.

(i) Singapore Standards

Standard	Subject
SS 71	Nomenclature of commercial timbers
SS 72	Treatment of timber and plywood with copper/chrome/arsenic wood preservative
SS 173	Glossary of terms relating to timber and woodwork
SS 332	Fire doors
SS 341	Safety glazing materials for use in buildings (human impact considerations)
SS	Timber doors, Code of Barrier Free Accessibility in Buildings

(ii) Other Standards

Standard	Subject
BS EN 438-2	High-pressure decorative laminate (HPL) sheets based on thermosetting resins. Determination of properties.
BS EN 485-3	Aluminium and aluminium alloys – sheet, strip and plate. Tolerances on shape and dimensions for hot-rolled products
BS EN 486	Aluminium and aluminium alloys. Extrusion ingots
BS EN 573-3	Aluminium and aluminium alloys - chemical composition and form of wrought products. Chemical composition

Standard	Subject
BS EN 755	Aluminium and aluminium alloys – Extruded precision profiles in alloys EN AW-6060 and EN AW-6063. Technical conditions for inspection and delivery
BS EN 1125	Panic exit devices operated by a lever, handle or push pad
BS EN 1154	Building hardware: Controlled closing devices.
BS EN 1155	Electrically powered hold open devices
BS EN 10088	Stainless steel-lists of stainless steels
BS EN 1303	Cylinders for locks
BS EN 10143	Continuously hot-dip coated steel sheet and strip – tolerances on dimensions and shape
BS EN 12020	Aluminium and aluminium alloys. Extruded precision profiles in alloys EN AW 6060 and EN AW 6063
BS 1161	Specification for aluminium alloy sections for structural purposes
BS 4255-1	Rubber used in preformed gaskets for weather exclusion from buildings. Specification for non-cellular gaskets
BS 4965	Decorative laminated plastic sheet veneered boards and panels
BS 5588	Fire precautions in the design, construction and use of buildings. Code of practice for shops, offices, industrial, storage and other similar buildings (AMD 9838) (AMD 10213)
BS 8118	Structural use of aluminium
BS 8300	Code of practice for the design of buildings and their approaches to meet the needs of disabled people
AS 1231	Aluminium and aluminium alloys. Anodised coatings for architectural applications
AS 3715	Metal finishing. Thermoset powder coating for aluminium and aluminium alloys

Standard	Subject
ASTM 2605-2	Wet applied organic coatings
ASTM 3715	Pressure sensitive tapes
ASTM C 509	Elastomeric cellular preformed gasket and sealing material
ASTM C 1172	Laminated architectural flat glass
ASTM D1056-00	Standard Specification for Flexible Cellular Materials– Sponge or Expanded Rubber

G8.3.3 Contractor's Submissions

- (a) The Contractor shall engage qualified and experienced personnel to carry out the work and submit necessary details for the doors and windows for the approval of S.O..
- (b) The Contractor shall submit the following items in 28 days advance for S.O.'s approval:
 - (i) Name of door and window manufacturer and copies of technical data relating to each product and certificates of compliance with SS 332 for all fire doors; and
 - (ii) Name of the ironmongery manufacturer/s and copies of technical details of all the ironmongery items. Wherever possible all ironmongery is to be obtained from the same manufacturer.

G8.3.4 Shop Drawing Submissions

- (a) The Contractor shall not commence fabrication until shop drawings have been reviewed and permission to proceed has been obtained from the S.O.. The Contractor to prepare and submit shop drawings to S.O. with inclusion of the following items:
 - (i) Elevation drawings of all doors;
 - (ii) Section showing door jamb, head, over panel, skirting, scale 1:2 for both fire and non-fire rated doors. Include door junctions for double doors;
 - (iii) Means of accommodating the specified acoustic standard;
 - (iv) A schedule of ironmongery for doors and windows in the format agreed, which shall include lock set details including security policies such as master key or common key requirements as shown in the drawings;
 - (v) Key schedule (format agreed with the S.O.); and
 - (vi) Layout drawings of all ironmongery

G8.3.5 Mock-ups

- (a) Prior to procurement, the Contractor shall provide necessary mock-up samples of doors and windows as required by S.O..

G8.3.6 Certification of Materials

- (a) The Contractor shall submit the certification from the manufacturer of the following materials/components, certifying that the respective material is of the correct grade, strength, size, finish, etc, and is in accordance with the relevant codes and standards specified:
 - (i) Doors;
 - (ii) Frames; and
 - (iii) Ironmongery.

G8.3.7 Test Reports

The Contractor shall submit the following to the S.O.:

- (a) Verification endorsed by the approved testing laboratories that the fire resistance and acoustic properties of the door sets (this includes the door in its frame with associated ironmongery) comply with specifications. Include the hinges, locks latches etc in place;
- (b) Test certificates and documents for installed fire doors; and
- (c) Documentation of endurance test for ironmongery.

G8.3.8 Handover of Keys of Completion

At completion of each phase (or part) of the Works, the Contractor shall:

- (a) Hand over three sets of all keys (or other quantity as specified by S.O.) each to be marked/labelled and provide a metal lockable key box to hold all keys.
- (b) Present confirmation from manufacturer of the number of keys supplied.
- (c) Maintain agreed security protocol for items such as master keys or common keys. These to be officially hand over without opportunity for deliberate or accidental breach of security.

G8.3.9 General Performance Requirements Dead Load

- (a) The self-weight and other associated dead loads of the complete door and frames are to be supported and transferred to the main building structure.

G8.3.10 Installation to Meet SCDF Requirements

- (a) The installation of fire rated doors shall satisfy the requirements of the Singapore Civil Defence Force - Fire Safety & Shelter Department (FSSD).

G8.3.11 Acoustic Integrity

- (a) The provision of sound insulated doors shall comply with the requirements of the local statutory authorities having jurisdiction over the Works.

G8.3.12 Glazing

- (a) The Contractor shall provide all glass and glazing including washing and cleaning down on completion when directed by the S.O..
- (b) The Contractor shall submit to the S.O. for approval, the name of the Glazier, before commencing any glazing.

- (c) The execution of glazing shall be taken as the Glazier's acceptance of the quality of frames to be glazed and Specifications for glazing. Any unsatisfactory conditions shall be reported to the S.O. prior to the commencement.

G8.3.13 Materials

- (a) Glass shall comply with BS 952, it shall be free from blemishes, bubbles, wares and other defects and shall be of the following types:
- (b) All 6 mm thick exterior glass shall have a reflective coating, be heat strengthened and comply with any architectural tints or finishes specified. Its properties shall consist of the following:

Visible Light Transmittance of	20%
Visible Light Reflectance in of	26%
Visible Light Reflectance out of	9%
Solar Energy Transmittance of	14%
Solar Energy Reflectance Outdoors of	8%
U-Value of	5.79 W/m ² k
Shading Coefficient of	0.39

- (c) 12 mm thick clear tempered glass shall exhibit the following properties:

Visible Light Transmittance of	85.8%
Visible Light Reflectance out of	7.7%
Visible Light Reflectance in of	7.7%
Solar Energy Transmittance of	70.1%
Solar Energy Reflectance out of	6.5%
U-Value of	5.62 W/m ² k
Shading Coefficient of	0.89

- (d) 15 mm thick clear tempered glass shall consist of the following properties:

Visible Light Transmittance of	84.3%
Visible Light Reflectance out of	7.6%
Visible Light Reflectance in of	7.6%
Solar Energy Transmittance of	65.9%
Solar Energy Reflectance out of	6.3%
U-Value of	5.65 W/m ² k
Shading Coefficient of	0.84

- (e) All heat strengthened & tempered glass shall comply with the requirements of ASTM C 1048-85.
- (f) Tempered glass shall comply with ANSI Z97.1-2004 for glazing materials & CPSC 16 CFR 1201 for safety glazing material.

- (g) All reflective laminated glass for the skylight system shall be 1G3.76 mm thick whose construction shall consist of an exterior lite of 6 mm thick heat strengthened glass and an interior lite of 6 mm thick heat strengthened glass with a 0.76 mm thick clearPVB (Polyvinyl Butyryl) interlayer.
- (h) Its properties shall consist of the following:

Visible Light Transmittance of	19%
Visible Light Reflectance in of	25%
Visible Light Reflectance out of	9%
Solar Energy Transmittance of	11%
Solar Energy Reflectance Outdoors of	8%
U-Value of	6.07 w/m ² k
Shading Coefficient of	0.38
- (i) All laminated glass shall comply with ANSI Z97.1-2004 and CPSC 16 CFR 1201.

G8.3.14 Floor Springs for Tempered Glass Doors

- (a) The Contractor is to supply and install double action floor springs specially developed for tempered glass doors. The floor springs shall comply with the following:
 - (i) Size to consist of length- 306 mm, width-108 mm, depth-40 mm;
 - (ii) Maximum opening angle of 130 degrees in each direction;
 - (iii) Closing from 130 degrees under total hydraulic control;
 - (iv) Thermo constant total mechanism is to be immersed in high quality hydraulic fluid to ensure smooth operating, even under differing temperature conditions;
 - (v) Equipped with pressure relief valve to protect mechanism if subjected to rough use;
 - (vi) A closing force of no less than 20 Nm at 2°;
 - (vii) Closer mechanism adjustment in cement box: Length-7 mm, Width- 10 mm, Height-4 mm; and
 - (viii) Full anti-corrosion treatment to mechanism and cement box. Cover plate to be in satin stainless steel
- (b) All glass louvres shall be of minimum 200 mm width to prevent easy breakage.
- (c) Putty for glazing shall conform to BS EN ISO 544, shall be designed for tropical use and shall receive one coat of approved paint 10 days after glazing.

G8.3.15 Samples and Handling

- (a) The Contractor shall submit samples of each type of glass to the S.O. for his prior approval before materials are ordered. The glass samples shall be not less than 150 x 150 mm in size.

- (b) The glass shall be delivered to Site in cut or stock sizes and in packages bearing the manufacturer's name and/or trade mark. The type, quality, thickness of the glass shall be clearly marked on the package.

G8.3.16 Installation

- (a) All glass fixed and glazing work shall be carried out in accordance with BS 6262.
- (b) Glazing beads, gaskets, glass adaptors and glazing sealants shall be neoprene. Glazing beads and other members shall be stiff enough and fixed at sufficient points to withstand a wind load arising from a design wind speed of 40 m/s.
- (c) Glass shall be accurately cut to fit rebates and carefully bedded, back puttied, sprung or pinned (beaded where specified) and evenly trimmed with all splayed edges straight and smooth. Defective, cracked or damaged glass shall be replaced.

G8.3.17 Mirrors

- (a) Mirrors shall be 3 mm thick approved best silvering quality, electro-copper backed with edges squared or bevelled and highly polished. Mirrors shall be fixed to timber frame as specified elsewhere.
- (b) Mirror glass shall be of silvering quality 6 mm thick float glass having a layer of silver and a layer of copper deposited on the back and covered with two coats of shellac varnish or other approved protective paint coating to resist moisture.
- (c) The size of mirrors shall be 600 x 300 mm for individual hand basins (without vanity slabs) and 750 mm high across the full width of vanity slab where hand basins are set into them.
- (d) Mirrors shall be fixed to masonry walls with waterproof plywood backing and dome headed screws complete with 12 mm natural anodised aluminium trimmings all round.

G8.3.18 Cleaning and Replacement

- (a) The Contractor shall allow for removing all cracked, damaged, scratched or otherwise defective glazing/mirror & leave sound on completion.
- (b) The Contractor shall allow for careful trimming and cleaning down all glass faces on completion.
- (c) Immediately before handing over, all glass and mirrors shall be cleaned, polished and replaced if damaged or broken. During the progress of the work, the glass shall be kept clean by prompt removal of foreign matter, particularly cement products.
- (d) Glass shall be cleaned with clean water with or without detergents added to it. Glass cleaning agents such as paraffin (kerosene), acid or abrasive cleaners which contain feldspar shall not be used for glass cleaning. Hard objects which may result in cuts, scratches or marks on the surface of the glass shall not be used for removing cement, paint or labels.
- (e) Glass and mirrors with cuts, scratches or marks on their surfaces shall be replaced at the Contractor's own cost.

G8.3.19 Aluminium Framed Windows and Doors

- (a) Extrusions
 - (i) Aluminium extrusions for door frames, window frames and panels and louvres, etc unless otherwise specified shall be constructed of extruded aluminium alloy or special profiled sections agreed by S.O..
 - (ii) Aluminium extrusions for door frames shall be 2.3 mm minimum wall thickness.
 - (iii) Aluminium extrusions for windows shall be not less than 1.2 mm and 1.8 mm thick for non-structural and structural members respectively.
 - (iv) All joints shall be electrically welded conforming to AWS structural welding code D1.1 or mechanically fastened. All fasteners shall be concealed.
 - (v) Platforms, webs, flanges, races and screw flutes of sufficient size shall satisfy all structural requirements and eliminate distortion of elements.

G8.3.20 Sheet, Strip and Plate

- (a) Aluminium louvered blades shall be of minimum 2 mm thick 'Z' profile.
- (b) Alloy shall be complying with BS EN 485-3.
- (c) It shall be not less than 1.6mm thick for hidden flashings, not less than 3mm thick for components exposed to view or to impact Stainless Steel Components.

G8.3.21 Finishes

- (a) The extrusions shall be finished in a polyvinylidene fluoride (PDVF) coating 70% 'Kynar 500' resin of colour to S.O.'s choice in 3 coats system. This consists of a factory applied thermo-cured inhibitive primer, a colour coat and a clear transparent top coat on properly pre-treated aluminium louvres. The coating shall meet or exceed the requirements as set out in AAMA-605.A (Voluntary Specification for High Performance Organic Coatings on Architectural Extrusions and Panels) or BS 4842 (Stoving Organic finishes on Aluminium Extrusions and Preformed Sections for External Architectural Application).
- (b) The panels must be factory protected with a peel-off PVC film after the application of the coating.
- (c) Perimeter trims of approved fluorocarbon coated aluminium shall be installed for all windows and louvres.

G8.3.22 Fixing of Glazing Panels

- (a) Glazing and fixing of glass shall conform to BS 6262. Performance of the window shall comply to SS 212 severe A for air infiltration and water penetration.
- (b) Window frames for fixed glass louvres shall have weathering jambs, heads and sill of minimum 102 mm x 45 mm with fixed clips to receive minimum 225 mm glass blades and with removable insect-proof screens in aluminium frames at the rear.
- (c) Window frames for fixed aluminium louvres shall have weathering jambs, heads and sills of minimum 102 x 45 mm to admit sufficient light to interior.

- (d) Screws, nuts, washers, bolts, rivets and other fastening devices incorporated in window units shall be of stainless steel, minimum grade 304 and shall be of sufficient strength to perform the functions for which they are required.
- (e) Glazing rebates shall be not less than 16 mm deep and sufficiently wide to provide adequate glass clearance and where glass dimensions exceed 650 mm, neoprene spacers are to be used.
- (f) Glazing beads shall be sufficiently heavy to maintain adequate pressure on spacers and shall be held in position by aluminium or stainless-steel screws or by approved snap-in type grooves, all with provision for adjustment.

G8.3.23 Stainless Steel Metal Panel Facing

- (a) Type 316 hairline finish, 1.2mm thick minimum conforming to BS EN ISO 18286, BS EN ISO 9445: Part 1 & Part 2, BS EN 10048, BS EN 10095, BS EN 10051 and BS EN ISO 18286, BS 7668, BS EN 10025: Part 1, Part 3 & Part 4, BS EN 10210: Part 1 laminated to 5-ply hardwood plywood of durability classification H per BS EN 636+A1 and DD CEN/TS 1099. Sand plywood smooth to remove surface irregularities prior to bonding plastic laminate face. The stainless steel facing is to be fully adhered to timber substrate. The door core is to be solid core. Provide hardwood lipping to all four sides of the door routed into the door core.
- (b) All external corners to be back cut with a maximum radius of 0.7mm. Locate joints in the metal sheet only as indicated on the drawings. All grain to be vertical.
- (c) Location: Refer Drawings.

G8.3.24 Ironmongery

- (a) Ironmongery shall be designed to perform the functions for which it is intended and shall be securely attached to the units. Unless otherwise specified, all ironmongery to be at least half-hour fire rated and suitable for external use.
- (b) All ironmongery for public communal areas and other specific locations is to comply with the Singapore Code on Barrier Free Accessibility in Buildings.
- (c) Ironmongery including door closer(s) to conform to requirements of endurance test as specified by selected manufacturer.
- (d) Component parts shall be of bronze, brass, stainless steel or other non-corrosive material compatible with the parent metal and of sufficient strength for the intended function. All items of ironmongery shall be subject to the approval of the S.O..
- (e) Provision shall be made for hinges, locks, catches, handles, door furniture, latches, opening, devices and pulls required for efficient use and operation.
- (f) Operating gear or mechanism to sashes shall be of such design as will adequately and easily control the airway of an opening sash by manual control within easy reach of the floor. Satisfactory hardware and operational gear shall be provided to perform the functions for which it is intended and in such a manner as will ensure adequate safety to the ordinary user in operation. Sashes shall not be removable from outside of the building.

- (g) Qualifications of Supplier: Recognised supplier of architectural finish hardware, with warehousing facilities, who has been supplying hardware for not less than 5 years, and who is, or who employs, architectural hardware consultant.

G8.3.25 Ironmongery - Products

- (a) Contractor shall provide end products of one manufacturer for each product in order to achieve standardisation for appearance, maintenance, and replacement. All finishes shall be grade 316 Stainless Steel.
- (b) Fasteners shall be stainless steel - grade 316.
- (c) All oil – impregnated ball bearing hinges shall be from one manufacturer and meet one of the standards listed:
- (i) ANSI A156.7 standard;
 - (ii) S 7352 / ISO 9227 class 4 corrosion resistance;
 - (iii) ISO 9001 and 14001 Certification;
 - (iv) All hinges shall be the same uniform five-knuckle design, available in low, medium and high frequency;
 - (v) Quantity per Door Leaf (Minimum):

<u>Door Height</u>	<u>Hinges</u>
Up to 1,500mm	1 pair
1,501mm to 2,100mm	1-1/2 pair
2,101mm to 3,000mm	2 pairs
3,001mm to 3,600mm	2-1/2 pairs
3,601mm and above	Subject to S.O.'s approval

- (vi) Hinge Height (Minimum):

<u>Door Width</u>	<u>Hinge Height</u>
Up to 915mm	115mm
940mm to 1,220mm	127mm
Over 1,220mm	152mm

- (vii) Width: Minimum for clearance of trim and 180-degree swing;
 - (viii) Exterior Hinges: Non-removable pin (NRP);
 - (ix) Joint Tolerance: 0.3mm maximum, gauged in CLOSED position;
 - (x) Finish: Satin stainless steel No. SUS 316L; and
 - (xi) Types and Manufacturers: As stated below or its approved equivalent.
- (d) Locks and Latch Sets
- (i) Mortise Locks: ANSI A156.13, Series 1000, Grade 1 and A115.1;
 - (ii) Materials: Brass or stainless steel;

- (iii) Trim: Wrought or forged lever handles and roses;
- (iv) Core Cylinders: Removal and interchangeable with other manufacturers industry standard, minimum of 6 pieces;
- (v) Bolt Throw: 16mm minimum;
- (vi) Lever Backset: 70mm;
- (vii) Finish: Satin stainless steel no. SUS 316L; and
- (viii) Keying:
 - 1) Lock Cylinders: Operate by Master Key system that allows a minimum of 400% vertical and horizontal expansion for future usage;
 - 2) Keylocks: Key new locks into existing Master Key system, as directed by S.O.;
 - 3) Keys: Three per lock; tag with schedule information; and
 - 4) Master Keys: Four; send by hand to S.O. by manufacturers only.
- (e) Construction Key System shall provide removable construction core system for locks.
- (f) Fire Exit Devices shall be provided at fire rated doors and shall comply with ANSI A156.3. Finishes shall be Satin chromium-plated No. 626.
- (g) Closers shall be sized in accordance with manufacturer's standards and shall comply with ANSI A156.4. Regular arm closures shall be mounted on pull side of doors. Parallel arm closures shall be mounted on push side of doors. On pair of doors provide closer on active leaf only, unless noted otherwise. Finishes shall be painted to match Door Frame Colour.
- (h) Push-Pulls:
 - (i) Solid metal, not plated;
 - (ii) Finish: Satin stainless steel No. 630, unless indicated otherwise;
 - (iii) Plates: Bevelled four edges, square corners;
 - (iv) Pulls: Bolted through door; and
 - (v) Push Plates: Countersink pull-through bolts and cover with push plate.
- (i) Stops and Holders:
 - (i) ANSI A156.16;
 - (ii) Machine Screws: In threaded anchors at concrete or masonry;
 - (iii) Self-Tapping Screws: At stud partitions, wood, or metal mountings; and
 - (iv) Finish: SUS 316 Stainless Steel.
- (j) Bolts:
 - (i) ANSI A156.16; and
 - (ii) Finish: SUS 316 Stainless Steel.

- (k) Kick Plates:
 - (i) Solid metal, not plated. Bevel four edges;
 - (ii) Width of door leaf less 40mm at single leaf and less 25mm at pairs; and
 - (iii) Finish: Satin stainless steel No. SUS 316.
- (l) Thresholds:
 - (i) Thresholds: One-piece full width of opening; extend beyond jamb where as shown on the Drawings;
 - (ii) Provide with stainless steel machine screws in threaded expansion anchors at concrete; and
 - (iii) Finish: Mill finish aluminium, unless indicated otherwise.
- (m) Weatherstrip:
 - (i) Finish: Clear anodised aluminium, unless indicated otherwise.
- (n) Silencers
 - (i) Product to S.O.'s approval;
 - (ii) At metal frame of each hinged door that does not have seals scheduled; and
 - (iii) Three at single leaves and two at pairs.
- (o) Templates
 - (i) Fabricate to template hardware applied to metal doors and frames;
 - (ii) Ensure that required templates are delivered to various manufacturers for fabrication purposes; and
 - (iii) Templates: Make available not more than 10 days after receipt of approved Hardware Schedule.
- (p) Exit Doors
 - (i) Exit Doors: Always openable from inside by simple turn of lever handle or push on panic bar without use of key or any special knowledge or effort, to include each leaf of door pairs.
- (q) Key Control System
 - (i) System:
 - 1) Expandable system with 150 key capacity expandable to 425 – Approximately 400% vertical & horizontal expansion.
 - (ii) Cabinet:
 - 1) Manufacturer's standard surface-mounted cabinet, single-door with continuous hinge, and pin tumbler lock, colour as selected from standard colours to match S.O.'s colour schedule.
 - 2) Complete system with standard components such as envelopes, transient key tags, reserve key tags, receipts and holders, and card index.

G8.3.26 Weather Proofing

- (a) All windows, sashes and other opening units shall have weather stripping of rubber gaskets, neoprene setting blocks and approved silicon water-resistant sealant.
- (b) Gaskets shall be extruded EPDM complying with provisions of BS 4255-1 shore hardness 35-45, Or Cellular rubber complying with ASTM C509. or Push-in gaskets i.e. Silicone free dry PVC push-in gaskets mitred at corners.
- (c) Doors shall have weather stripping of certified high pile wool mohair with silicon water-resistant treatment to eliminate water leakage in extreme weather conditions. All joints between aluminium head or cill and concrete surfaces shall be also similarly treated with silicon water-resistant sealer. The completed works is subjected to waterproofing test against weathering.

G8.3.27 Air Tightness and Leakage

- (a) All windows on the building envelope shall not exceed the air leakage rates specified in SS 212- Specification for Aluminium Alloy Windows.

G8.3.28 Installation

- (a) The provision of trimming members shall be incorporated in the design and construction of units as necessary to suit the architectural requirements specified.
- (b) Expansion and construction of the member of frame must be able to take place freely. Under no circumstances will the provision for expansion be allowed by slotting the fixing holes in brackets to allow sliding movement between the brackets and the building structures.
- (c) Wherever possible all metal shall be formed, cut, drilled, tapped, welded, fitted or otherwise fabricated in the shop. Fixing lugs of all metal sub-frames shall be suitably galvanised or primed with zinc chromate before building in.
- (d) The units shall be set plumb, square and level and shall be built into openings prepared for this purpose and shall be securely held in correct vertical and horizontal alignment.

G8.3.29 Warranty

- (a) The Contractor and his specialist sub-contractor shall jointly furnish a ten (10) years warranty for all aluminium works in an approved form, against faulty design, materials and workmanship to the Employer and the S.O. upon handover of the Works.

G8.3.30 Security Grilles to Windows

- (a) Security grilles shall be provided where specified and to all store rooms and workshops/stores of all buildings. Diamond profile solid mesh aluminium grilles shall be fixed to aluminium framed glass windows and louvres and shall be incorporated to the external faces of top hung/casement windows. For louvre windows, they shall be incorporated between the glass louvres and the removable insect screens with matching aluminium spacers and frames for fixing.

- (b) The design and details of incorporation and fixing of security grilles shall be submitted to the S.O. for approval before manufacturing and incorporation.

G8.3.31 Sliding Folding Doors

- (a) Sliding folding doors shall be manufactured out of mild steel and shall be galvanised and powder coated finish to S.O. approved colour to suit the individual building.
- (b) Sliding folding doors shall be of the top hung double leaf type with odd numbers of folds of 300 mm to 350 mm to each leaf with overhead rollers and bottom guides. Doors shall operate manually with smooth action and be held in position by a door retainer when open. Vertical and horizontal pelmets shall be provided to conceal the rollers and folds when the doors are opened. Mild steel panels used for the doors shall not be less than 16 SWG thick and shall be welded onto a rigid frame together with all brackets, hinges, handles, locks, etc. Three (3) keys shall be provided with the locks. The hinges, handles and locks shall be of stainless steel and all other metal components shall be heavy duty galvanised.

G8.3.32 Roller Shutter Doors

- (a) Roller shutter doors shall be manufactured out of mild steel and shall be hot dip galvanised and powder coated finish to a colour approved by the S.O. to suit the individual building.

G8.3.33 Regulatory Requirements

- (a) Fire-Rated Assemblies:
 - (i) Conform to Singapore codes for fire-rated openings;
 - (ii) Install door and frame assemblies to conform to Singapore codes for 2-hour fire-rated class indicated;
 - (iii) Fire door assemblies, including products requiring electrical connection, shall be tested and labelled by an independent agency approved by authorities having jurisdiction; and
 - (iv) Fire-rated overhead coiling door to comply with Singapore Standard SS 489-Fire Shutters.

G8.3.34 Drum Assembly

- (a) The drum assembly shall be an extremely rigid lateral torque-tube consisting of a 200 mm dia. mild steel tube sheathed over a complete length of 38 mm dia. mild steel solid axle supported by suitably-spaced cast iron drum wheels. The counterbalance springs shall be manufactured from 8 mm to 10 mm dia. wires of various lengths and coildiameters and shall have one end secured to the axle and the other secured to the drum wheel so that spring torsion is applied as the tube rotates calculated in accordance with the width, height and weight of each door to achieve the torque required for optimal counterbalance.

G8.3.35 Door Curtain

- (a) The door curtain shall consist of 100 mm x 1.2 mm thick galvanised steel scroll-profiled interlocking slats with proprietary nylon end-clips secured to alternate slats to prevent lateral movements. Curtain penetration into the guide channels shall not be less than 100 mm on each side. The bottom rail shall be a single-length inverted 'T' steel section of 5 mm thickness, with the base of the rail containing a channel cavity to retain a seamless profiled PDM weather seal.

G8.3.36 Door Guide

- (a) The guide channels shall be heavy-duty galvanised steel profiles of overall dimensions of 120 mm x 100 mm x 5 mm thick, incorporating full-height seamless rubber strips at the lips for smoother and quieter operation. It shall be curved at the top for curtain entry and shall have removable back covers for easy inspection and maintenance. The guide channels shall also incorporate a mini current-collector strip to supply to the safety-edge mechanism, thereby eliminating the need for a trailing exposed spiral cable.

G8.3.37 Door Operation

- (a) The drive unit shall consist of a suitably-sized 3-phase continuous-rated electric motor driving a totally enclosed reduction gearbox with a reduction of 45 to 1. It shall have a chain-operated manual sheave wheel which automatically engages whenever the hand chain is pulled and disengages whenever the door is being electrically operated so that the hand chain does not travel with the motor.
- (b) The motors control shall incorporate an adjustable torque limiter which prevents the door from overdriving should the limit switch fail or become mis-positioned. The control box shall be a reversing contactor with a built-in thermal overload feature. It shall have 'Up', 'Down' and 'Stop' push-buttons with a choice of 'hand-on' or 'hands-off' operation. The door operation shall automatically stop at the desired upper and lower limits via adjustable limit switches.
- (c) The door shall have a bottom safety-edge designed to immediately stop a closing sequence upon contact with an obstruction. This shall consist of a low-voltage resistance-activated contact strip incorporated into the bottom rail weather seal. It must be capable of being activated with 'finger-pressure' sensitivity.

G8.3.38 Locks

- (a) Locks shall be proprietary waist-high locks with lock bars locking into door guides. Locks shall have finger-grip recesses for lifting or closing door. Locksets shall be secured from inside and shall not have fasteners visible from the outside. Dual-mode operation shall permit the option of lock operation from inside without the use of key. Doors with a width exceeding 3.5 m shall be fitted with two locksets, one on each side of the door.

G8.3.39 Box Housing

- (a) Box housings shall be provided for each roller shutter door, suitable for mounting internally.

G8.3.40 Fire Rated Glazing Panels and Glass Doors

- (a) The Contractor shall provide fire rated glazing panels and glass doors to the specified fire rating.
- (b) All glass used shall be multi-laminated clear safety glass with interlayers of sodium silicate sheet which would intumesce upon intense heating. The thicknesses to conform to the fire rating requirements specified.
 - (i) Safety glass used shall comply with the following specifications:
 - 1) Mechanical impact of 600N to BS 6206 Classification A for safety glass;
 - 2) Resistance to shattering by puncturing AS 2208
 - 3) Fire Resistance for stability, integrity and insulation BS476 pt 8:1972
 - 4) For fire resistance ISO 834
 - 5) Stability, integrity and insulation SS 332 : 2007
 - 6) Cyclic movement SS 332 : 2007
 - 7) Code for Fire Precautions for Buildings, 1991-Fire Doors Table 3.4A

G8.3.41 One-Hour Rated Glass Doors

- (a) Glazing shall be made up of 5 layers of safety glass laminated with 4 layers of sodium silicate interlayer which would intumesce upon intense heat. Total glass thickness is 21 mm thick.

G8.3.42 One And A Half/Two Hours Rated Glass Door

- (a) Glazing shall be made up of one piece of laminated clear safety glass, 17 mm thick and bonded to a piece of 5 mm thick ceramic heat resistant glass, with an air space of 4-6 mm in between. The ceramic heat resistant glass is laminated with a toughened transparent 0.2 mm thick PVB membrane to achieve impact resistance, in accordance with BS 6206. Total glass thickness is 27 mm

G8.3.43 Frame Construction

- (a) The composite frame shall consist of an internal structural steel frame insulated with 20 mm thick calcium silicate boards & capped with a steel frame clad in stainless steel finish.
- (b) The glass is to be held by the insulated framing which is then capped all round by 1.5 mm thick steel plate to give impact resistance compatible to the glass itself. Total frame cross section shall be 68 mm thick and frame width shall be 85 mm wide.
- (c) The gaps between the frame are to be protected against heat & smoke infiltration by an intumescent seal.

G8.3.44 Floor Spring

- (a) The Contractor shall provide heavy duty hydraulic floor springs for left & right pivoted doors. These shall be designed as a single action, handed floor spring in accordance with DIN 18263, Part 3 for application on fire & smoke-check doors and shall comply with the following Specifications:
- (i) The spring strength shall be determined by the door leaf width:
- | Door leaf width (mm) | Spring Size | Min torque in Nm at a door opening of 2 |
|----------------------|-------------|---|
| Up to 1100 | 3 | 26 |
| 1100 to 1250 | 4 | 43 |
| 1250 to 1400 | 5 | 63 |
- (ii) Two speed hydraulically controlled closing from a full 180 degrees of door opening.
- (iii) Adjustable latch speed from 7 of door opening.
- (iv) Built-in mechanical back check approximately 85 and 165.
- (v) Thermostatic control valves to ensure constant closing speed under varying temperatures. Mechanism is to be totally immersed in hydraulic fluid.
- (vi) Equipped with pressure relief valve to prevent damage to the closer from forced closing.
- (vii) Closer mechanism adjustment in cement box: Length-6 mm, Width- 6 mm, Height-4 mm.
- (viii) Interchangeable spindles to suit varying clearances between door and closer mechanism.
- (ix) Full anti-corrosion treatment to mechanism and cement box, Cover plate is to be in satin stainless steel.

G8.3.45 Fittings

- (a) Handle shall be a horizontal Dee shaped handle in stainless steel, subject to the approval of the S.O..
- (b) Lock system shall consist of a concealed pivoted door lock system to the top of the frame.

G8.3.46 Fire Rated Glass Viewing Panels

- (a) All glass used shall be multi-laminated clear safety glass with interlayers of sodium silicate sheet which will intumesce upon intense heating. The thickness shall depend on the fire rating requirement specified as follows:
- i) Half (1/2) hour rated glass : 12mm thick multi-laminated glass panel
- ii) One (1) hour rated glass : 17mm thick multi-laminated glass panel
- iii) One and a Half (1:1/2) hour rate glass panel : 19mm thick multi-laminated glass

- | | | |
|-----|-------------------------------|--|
| iv) | Two (2) hour rate glass panel | 12mm thick multi-laminated glass bonded to a 5mm thick ceramic heat resistant glass with a 4-5 mm air space in between, making a total thickness of 27mm |
|-----|-------------------------------|--|

- (b) The composite frame construction is to be similar to Item G3.7.7.1.

G8.3.47 Steel Doors and Frames

(a) General

(i) Standard Steel Doors Product Description:

- 1) Exterior Doors (Insulated): ANSI A250.8, SDI 108, 44.45 mm thick.
 - Level 1: Standard duty, Model 2, seamless design.
- 2) Interior Doors (Nonrated): ANSI A250.8, SDI 108, 44.45 mm thick.
 - Level 1: Standard duty, Model 2, seamless design.
- 3) Interior Doors (Fire Rated): ANSI A250.8, SDI 108, 44.45 mm thick.
 - Level 1: Standard duty, Model 2, seamless design.

(ii) Standard Door Frame Product Description:

- 1) Exterior Frames:
 - Level 2 for Door Models 2, nominal 16-gauge/1.35-mm-thick material, base metal thickness.
- 2) Interior Frames:
 - Level 2 for Door Models 2, nominal 16-gauge/1.35-mm-thick material, base metal thickness.

(b) Materials

- (i) Frame, Door, Panel, Louver, and Glazing Bead Material: ASTM A1008, cold-rolled steel.
- (ii) Frame Reinforcement: ASTM A653 sheet steel with 35.44 gm-per-square-meter galvanized coating.
- (iii) Door and Panel Core Construction: steel channel grid, with mineral fiberboard insulation in fire-rated doors; in accordance with listed construction.
- (iv) Door and Panel Acoustic Materials: glass fiber insulation, or manufacturer's standard insulation as required to achieve specified thermal and acoustic performance.
- (v) Hardware Reinforcement: ANSI A250.6 and for fire-rated openings, SS332:2000.
- (vi) Glass for Door Lights: as specified in Section 08800, Glazing. Frame should be 45 degree pitched for dust proof.
- (vii) Protective Coating: fibered asphalt emulsion.

- (viii) Primer: rust-inhibiting type compatible with finish coating specified in Section 09915, Building Painting.
- (c) Fabrication
 - (i) Fabricate frames and assemble as a shop-welded unit with mitered corners. Grind welds smooth and apply filler; sand smooth.
 - (ii) Size units to allow for installation clearances, shim spacing at perimeter of installed assembly, and deflection of lintel.
 - (iii) Provide flush, hairline, and weatherproof joints and connections. Match components for continuity of line and design.
 - (iv) Transom Bars for Glazed Lights: fixed type of same profiles as jamb and head.
 - (v) Fabricate double doors with steel T-shaped astragals.
 - (vi) Fabricate frames and doors with hardware reinforcement plates welded in place. Locate reinforcement in accordance with hardware location templates provided for door hardware specified in Section 08710, Door Hardware.
 - (vii) Reinforce frames wider than 1200 mm with roll-formed steel channels fitted tightly into frame head, flush with top.
 - (viii) Weld steel mortar guard boxes to frames protecting hardware reinforcement and cutouts from grout intrusion on door frames that are grouted full.
 - (ix) Prepare frame for silencers. Install three single silencers for single doors and mullions of double doors on strike side and two single silencers on frame head at double doors without mullions.
 - (x) Attach fire-rated label to each frame panel and door unit of fire-rated doors and frames.
 - (xi) Close top edge of exterior door and doors in wet interior areas and cleanroom environments with inverted steel channel closure. Seal joints watertight.
 - (xii) Unless otherwise detailed, fabricate frames with 50.8 mm face profile and 12.7 mm return to wall surface.
 - (xiii) Fabricate frames for masonry wall coursing with head member 100 mm head.
 - (xiv) Door Top Rail: 150 mm nominal.
 - (xv) Door Bottom Rail: 406 mm.
 - (xvi) Fabricate door lock and hinge edge rail with seam wire welded full length and ground smooth.
 - (xvii) Fabricate glazed openings in the door with manufacturer's standard glass molding. Place glazing stops on secure side of glazed frames. Fasteners shall be stainless steel.

- (xviii) Install the trim into the door as a four-sided welded assembly. Fit the trim into a formed area of the door face; flush with the door face; and interlocked into the recessed area. Miter, reinforce, weld, and fill smooth the corners of the assembly. Use the same trim on both sides of the door unless scheduled or detailed otherwise. Conceal fasteners. Use the same trim for label and non-label doors.
- (xix) Flush panel: Form to thickness and dimensions shown and comply with applicable requirements for flush doors. Prepare panels for concealed support and anchorage.
- (xx) Provide junction box in door frames scheduled to receive electric hinge power transfer, electric strike, smoke closer and door monitor switches. Acceptable product: Stanley JB-1/ JB-2 or equivalent approved by the Owner.
- (d) Accessories
 - (i) Louvers: door manufacturer's standard roll-formed steel, slat blade design, minimum 16 gauge/1.35 mm, with vertical reinforcement at 300-mm centers and minimum 50 percent free area. Flush-style frame with flush surface fasteners. Finish as specified and scheduled for doors.
 - (ii) Grout: maximum 100 mm slump Portland cement grout.
 - (iii) Jamb Anchors: manufacturer's standard as appropriate for indicated substrate.
 - (iv) Silencers: resilient rubber.
 - (v) Glazing Bars: rolled steel square channel shape; square corners; prepared for countersink style tamperproof stainless steel screws.
 - (vi) Shim Space Insulation: loose glass fiber type.
- (e) Finishes
 - (i) Exterior Door and Frame Assemblies: ASTM A653, A60, G165.
 - (ii) Interior Door and Frame Assemblies Subject to Wet or Corrosive Environments: ASTM A653, G60.
 - (iii) Interior Door and Frame Assemblies Subject to Normal, Dry Environments: manufacturer's standard rust-inhibitive primer compatible with finish paint coats.
 - (iv) Phosphate treat for paint adhesion.
 - (v) Finish: factory finished thermosetting epoxy, color as selected by Owner.
- (f) Color
 - (i) Exterior Door and Frame: Color to match adjacent exterior wall paint.
 - (ii) Interior Door and Frame: as indicated in Door Schedule on Drawings.
- (g) Door Frames Installed in Masonry or Concrete Walls: Fill mortar inside of jamb box.

G8.3.48 Timber Doors and Frames

- (a) Material requirements shall be as follows:
 - (i) Unless otherwise specified all timber shall be Chengal, Merbau or Balau.
 - (ii) Moisture content when fitted shall be between 10-15%, moisture content between adjacent timbers to be no more than 3%.
 - (iii) Pressure impregnate all timber, impregnation system to approval of S.O..
 - (iv) Fire retardant treatment system, to be submitted to the S.O. Season and kiln dry timber after treatment.

G8.3.49 Timber Doors

- (a) Door frames, unless otherwise indicated shall be of Chengai or Merbau, sized 150 x 64 mm rebated as shown to form doorstops.
- (b) Doors shall be of type, size and thickness as indicated on the Drawings and shall be of first quality construction to B.S. 459. Doors shall be obtained from one manufacturer as approved by the S.O. and shall be guaranteed against any warp, wind or other defects, other than fair wear and tear. Should any defect or defects become apparent during the Defects Liability Period, the affected door shall be replaced and finished to approval at no extra cost.
- (c) Extra sizes of doors shall be obtained from the job; nominal minimum sizes only are indicated on the Drawings.
- (d) Unless otherwise specified, all meeting stiles of paired doors shall be rebated.

G8.3.50 Timber Frame/Architraves

- (a) Architraves shall be provided and fixed to timber door frames. They shall be of hardwood size 19 x 12 mm mitred at corners.
- (b) Where door jambs are flush with the face of brickwork, architraves shall be splayed externally 38 x 18 mm mitred at corners.

G8.3.51 Door Furniture

- (a) Door furniture shall be to the approval of the S.O. and shall be fitted with matching screws. Door furniture requirements shall be confirmed by the S.O. on Site.
- (b) The Contractor shall provide the door furniture as loose items to the approved door manufacturer(s) who shall install the door furniture into the doors.

- (c) The furniture shall generally be as follows:
- i) External single doors : Double cylinder upright night latch with mortice lever handle.
 - ii) External double doors : Leading leaf as for single door. Following leaf to have 200mm flush bolts tip and bottom.
 - iii) Internal single doors : Cylindrical upright mortice lock, key outside, thumb-turn inside with knob handle.
 - iv) W.C. doors : Aluminium indicating privacy lock. Hinges are to be self opening of an approved pattern.
 - v) Door closers doors : To be fixed to all single leaf (except toilet doors) and leading leaf of double doors.
 - vii) Internal doors kicking : To have a stainless steel plate and 225mm x 75mm stainless steel finger plate on each side of each leaf.
- (d) Handles shall be horizontal Dee shaped handle in stainless steel, subject to the approval of the S.O..
- (e) Flush bolts for the tops of doors shall be provided with suitable top plates, and those for the bottom of doors shall be supplied with cleanable socket.
- (f) All external and internal door furniture shall be stainless steel with a satin polished finish except for the barrel bolts which shall be of brass.

G8.3.52 Facing

- (a) Stainless Steel Facing shall conform to BS EN 10088 Grade 1.4401, and 1.5mm thick conforming to BS 1449-2.
- (b) Plastic Laminate Facing shall be 1.4 mm thick durability classified D4, performance rated heavy duty (HD) to BS 4965 and BS EN 438-2.
- (c) Plastic laminate to be warranted against UV deterioration by the manufacturer.

G8.3.53 Locks

- (a) Locks shall be supplied with three (3) individual keys. In addition, four (4) Simple Master Keys shall also be supplied. All locks required for fire doors, rolling shutters, grilles, steel doors, gates and the like shall be incorporated in the master key system.
- (b) Each lock shall be supplied with three (3) keys and shall be incorporated in two (2) master key systems, one system for all external doors and another system for the internal doors. Four (4) master keys shall be supplied for each system.

G8.3.54 Fire Doors

- (a) All fire doors whether of steel or timber shall conform to SS 332.
- (b) All such doors shall be tested and labelled before delivery to site. Doors and frames are to bear a certification label from the relevant local authority and a certificate of conformance is required from the manufacturer for each type of fire door.
- (c) All fire doors shall be fabricated and installed to specialist's standards, specifications and instructions and to suit door opening size, profile and appearance of door type as shown on the drawings.
- (d) The Contractor shall note that the price for all steel and timber fire doors shall include the door head and jambs, grouting behind the back of frames in the case of metal jambs and three numbers of galvanised steel anchors to each door jamb and two numbers to the door head. Ironmongery and door accessories, such as door closer(s) shall be compatible with the fire rating of the door.
- (e) Glazing components for vision panel in Fire Doors to comply with standards of SS 332. Glazing components to conform to BS 5588, surface spread of flame to class 1 BS 476.
- (f) Materials used for fire doors shall be identical with the door system described in the test certificates. All fire doors are to be approved and of construction identical to door system submitted for fire resistance tests.
- (g) Changes to the core material or approved thickness are not allowed. Any variation /substitution in the door and frame size, construction material, ironmongery, and method of fixing/installation are to be subject to further approval by the Regulatory Authorities.
- (h) All fire doors shall be self-closing and shall match the finish of other adjoining doors around the vicinity and their frames shall be finished in the same colour or colours unless otherwise specified or as approved by the S.O..
- (i) Where fire rated doors are set below suspended ceilings or above raised floors include for infilling void above or below door with fire rated infill of the same rating as the door. The construction of the infill is to be to the approval of the S.O..

G8.3.55 Hinged Fire Timber Door of Half Hour Rating

- (a) Half hour hinged fire doors shall be approved type with nominal thickness of 45 mm.
- (b) All door leaves shall consist of high density Fire Research strandboard core and faced on each side with MDF board.
- (c) Single and double leaf doors shall have internal Solid timber edge strips fitted to both stiles with intumescent strips bonded internally to the sides and top edge of the door leaf.
- (d) Where overhead transom panels are required, all transom panels shall be of similar construction to the door leaf and shall be fitted between a high density solid timber transome and head of the door frame above the door leaf.

- (e) All fire doors shall be approved by the Fire Safety Bureau and shall conform to the Singapore Productivity & Standards Board's requirements and be labelled with a certificate of approval. The doors shall come complete with the necessary door frame and ironmongery, subject to the S.O.'s approval.
- (f) ½ hour door frames shall be approved type as shown on the drawings. All frames shall be made from high density solid timber rebated and to meet fire rating requirements.
- (g) All frames shall be appropriately prepared to receive door closer arms, lock striker plate and hinges. A timber spreader is to be fitted to the bottom of each frame and can be fixed or removed after installation subject to project requirements.
- (h) All frames shall be installed true and plumb into door openings by means of steel bolts located at intervals along both jambs and at the head in the case on installing in masonry walls.
- (i) The number of steel bolts used is subject to the height of each door frame and shall follow the manufacturer's installation specification.

G8.3.56 Hinged Fire Timber Door of 1, 2, 3 and 4-hour Ratings

- (a) Unless otherwise specified, fire timber doors of one-hour fire ratings and above shall be constructed of minimum 1.22 mm (18g) thick zinc coated steel sheet press formed to provide a fully flush, double skin door shell with lock seam joints at stile edges. The internal construction of the door shall comprise minimum 1.6 mm thick stiles, top and bottom rails and reinforced with minimum four number intermediate 16g channels and doors shall be infilled with fire-resisting materials. Mere compliance with this specification shall not relieve the Contractor from his liabilities should the doors constructed fail to achieve the mentioned fire ratings.
- (b) Hinged fire doors shall be of 1, 2, 3 and 4-hour ratings having a nominal thickness of 45 mm. The fire doors shall conform to ISO 3009-2003, and SS 332 and shall be manufactured by approved manufacturer.
- (c) All fire doors shall be approved by the Fire Safety Bureau and shall conform to the Singapore Productivity & Standards Board's requirements and be labelled with a certificate of approval. The doors shall come complete with the necessary door frame and ironmongery, subject to the S.O.'s approval
- (d) All door leaves shall consist of Fire Research solid moulded refractory core and strengthened with galvanised steel angle sections at the hinge, latch and closer positions.
- (e) Single and double leaf doors shall have internal edge strips made of calcium silicate board, filled to both stiles and intumescent strips bonded to the sides and top edge of the door leaf.
- (f) All finishes shall be of first grade fire retardant treated plywood, laminate sheet or teak plywood for 1, 2 and 3-hour doors and steel sheeted for 4-hour doors.
- (g) Plywood, teak plywood or laminate sheet finish shall weigh approx. 30/m². Mild steel sheet shall weigh approx. 45 kg/m².

- (h) Fire door frames shall be approved double rebated type having a girth of 300 mm. All frames shall be accurately formed from 1.5 mm thick mild steel sheet and shall be 150 mm wide with a rebated depth of 25mm to meet fire rating requirements.
- (i) All frames shall be reinforced to receive door closer arms, and suitable lock striker fixing plate with mortar guard to be fitted to the jamb in the case of single leaf door. Steel spreader is to be fitted to the bottom of, each door frame and can be fixed or removed after installation subject to project requirements.
- (j) All fire door frames shall be installed true and plumb into door openings and shall be grouted solid with cement mortar. All frame shall be provided with “fishtail” cross brackets to be built into mortar joints to at least 75 mm in depth at vertical spacings along the jambs.
- (k) The number of cross brackets used is subject to the height of each door frame and shall follow the original manufacturer’s installation instruction.

G8.3.57 Sliding and Folding Partition Systems

- (a) Where specified or shown on the Drawings the Contractor shall provide approved sliding and folding partitions. The partition system shall be able to achieve sound insulation value of R'W 30db. It shall be constructed of dual layer, construction panels made of 13 mm three ply quality heavy non-inflammable chipboard with a level spring to ensure a totally flat surface and stability when the partition is closed.
- (b) The surface shall be laminated with high pressure laminates which have a minimum thickness of not less than 1.2 mm.
- (c) Materials used to construct the sliding, folding partition system must be extremely resistant to scratches and impact and be immune to organic acids.
- (d) The heavy non-inflammable chipboard panels shall be tested in accordance with Din 4102, BI or BS, and shall be Class O Density: Approximate 610- 880kg Per Cubic Metre.
- (e) The width of each folded panel shall be 560 mm.
- (f) Whenever required, an extension hanging structure shall be provided which shall have been designed by a qualified engineer. No extra cost will be entertained in this respect.

G8.3.58 Sliding and Folding Gear

- (a) The partitions shall be hung centrally under the lintel and fold to both sides. The folding units should consist of an odd number of full door leaves with one half leave hinged to a jamb.

- (b) The sliding and folding gear shall comply with the following:
 - i) Trade : Cold rolled steel section hot dip galvanised which must be greased after installation.
 - ii) Brackets : Cast aluminium or pressed steel.
 - iii) Hangers : Fitted with maintenance free precision ball bearings sealed for life or integrally moulded plain bearings.
 - iv) Steel wheels & other parts : Zinc plated and adjustable in height.
 - v) Hinges : Aluminium alloy satin anodised finish.
 - vii) Flush pulls and bolts : Satin anodized aluminium.
- (c) Floor runner(s) shall not be used and the operation shall be feather light and noiseless. The whole partition system shall be complete with all necessary accessories and ironmongery.

G8.3.59 Bullet Resistant Fixed Glazing Systems

- (a) The system shall feature ballistic resistant aluminium frame construction, easily replaceable laminate material, replaceable surface protective film laminate, it shall be non-spalling, multiple impact, shall not spider, crack or whiteout on impact and shall provide vision after ballistic or physical attack and shall come delivered assembled, ready for installation.
- (b) The integrated Ballistic Resistant Fixed Glazing Window System (including the frame) shall be an integral design system by specialist contractor. Any proposal that does not meet this requirement will be rejected.
- (c) All ballistic resistant Aluminium extruded framing are to be constructed to defeat 0.38 Super Automatic rounds. (3 shots)
- (d) The installer shall be the authorized and approved installer of the manufacturer for the intended integrated ballistic resistant fixed glazing window system and the installer shall present a copy of their "Certificate of Registration" certifying their Quality System is in accordance with, and functions as required under ISO 9001:2000.
- (e) Applicable Standards
 - (i) HPW-TP-0500.00;
 - (ii) ASTM F1233-93;
 - (iii) L Standard 752; and
 - (iv) ISO 9001:2000.
- (f) The integrated ballistic resistant fixed glazing window system and its framing shall have a warranty for one (1) year against manufacturer's defects in material and workmanship from the date of installation. The Glazing shall be warranted for a period of five (5) years from yellowing.

G8.3.60 Bullet Resistant Glazing

- (a) Glazing material shall be LAMINATE (minimum 4-ply polycarbonate and acrylic laminate) meeting the requirements
 - (i) Underwriters Laboratories UL Standard 752 Level 6 (high velocity 9mm submachine-gun) and Supplementary Shotgun (12 gauge rifled lead slug and lead buckshot). As tested by HP White Laboratories;
 - (ii) European Union Standard prEN1063:1933 (draft) Class B4 (.44 Magnum, .357 Magnum, 9mm parabellum, and .22 long rifle) and Class SG2 (12 gauge shotgun Brenneke solid lead slug). As Tested by the German State Institute;
 - (iii) German Standard DIN 52290 Part 1 Class C3 SF (.44 Magnum). As tested by German State Institute;
 - (iv) Bullet Resistant Glazing Testing requirements of the Royal Hong Kong Police Force Crime Prevention Bureau (Ref. DID/CPB 5/2), using Type 54 Chinese Military Pistol (7.62mm) and Type 51-1 solid steel, Banded, ammunition (3 rounds). As tested by HP White Laboratories;
 - (v) Flexural Strength not less than 13,500 psi (ASTM D790);
 - (vi) Light Transmission (ASTM D1003) of 81%;
 - (vii) Non-spalling, multiple impact, will not spider, crack or whiteout on impact and provide vision after ballistic or physical attack;
 - (viii) Replaceable surface protective film laminate for ease of maintenance; and
 - (ix) Material shall be a total thickness of 1.300" + 5% and weighs 8.1 lbs per sq. ft. or less.

G8.3.61 Hollow Metal Bullet and Forced Entry Resistant Door Systems

- (a) The system shall feature a vision panel with laminate material as defined, a reinforced frame complete hardware outfitting. The door and frame shall be delivered as a complete unit
- (b) The hollow metal bullet & forced entry resisting door system shall be designed to defeat UL 752 Level 3 ballistic assaults. It shall consist of continuous gear hinge and shall be an interior/exterior hollow metal swing.
- (c) The hollow metal bullet and forced entry resistant door System (including the frame) shall be an integral design system by specialist contractor. Any proposal that does not meet this requirement will be rejected.
- (d) The installer shall be the authorized and approved installer of the manufacturer for the intended hollow metal bullet and forced entry resistant door System. And the installer shall present a copy of their "Certificate of Registration" certifying their Quality System is in accordance with, and functions as required under ISO 9001:2000.
- (e) Applicable Standards
 - (i) HPW-TP-0500.00;
 - (ii) ASTM F1233-93;

- (iii) UL Standard 752; and
- (iv) (d) ISO 9001:2000.
- (f) Warranty
 - (i) The hollow metal bullet and forced entry resistant door System shall have a warranty for one (1) year against manufacturer's defects in material and workmanship from the date of installation. The Glazing shall be warranted for a period of five (5) years from yellowing.

G8.3.62 Bomb Blast Restraint System for Window and Glass Panels

- (a) The system shall feature a bomb blast restraint system, mechanical anchoring assembly, fragment retention laminate, glass strengthening treatment and shall be aesthetically acceptable and blend with the primary window frames.
- (b) The Bomb Blast Restraint System shall comply with US General Services Administration GSA Level 2 in accordance to ASTM 1642-96. As tested by Applied Research Associates (ARA). The system shall meet:
 - (i) GSA Level 2 Protection Level Very High & Hazard Level None; and
 - (ii) No significant fragments; Dusting or very small fragments near sill or on floor acceptable.
- (c) The system shall be an integrated design system to be installed by qualified installer that have been trained and approved by the manufacturer for the intended security system. The Qualified installer shall present a copy of their Certificate of Registration certifying their Quality System is in accordance with, and functions as required under ISO 9001:2000.
- (d) Applicable Standards
 - (i) US General Services Administration (GSA) Standard Test Method for Glazing and Glazing Systems Subject to Air blast Loadings;
 - (ii) ASTM 1642-96; and
 - (iii) SB SS341:2001.
- (e) Window System shall be warranted from defects in workmanship and quality for a period of one (1) year from practical completion. Laminate shall be warranted for a period of five (5) years against yellowing, coating failure and delamination.

G8.3.63 Mechanical Anchoring Assembly

- (a) Anchoring extrusions shall be fabricated from 6063-T5 tempered aluminium alloy.
- (b) Threaded fasteners shall be stainless steel or zinc-plated steel in accordance with ASTM A164. Fasteners shall be installed every six (6) inches.
- (c) The adhesive tapes or transfer adhesives used shall have the following properties:
 - (i) Minimum tensile strength of 200 pounds per square inch (psi) when tested in accordance with ASTM D412;

- (ii) Minimum peel adhesion of 15 pounds per lineal inch (pli) when tested in accordance with ASTM D333;
 - (iii) Minimum cleavage peel strength of 30 pounds per lineal inch (pli) when tested in accordance with ASTM D1062;
 - (iv) Minimum dynamic shear strength of 90 pounds per square inch (psi) when tested in accordance with ASTM 3163;
 - (v) Moisture resistant; and
 - (vi) Maintains adhesion at sub-zero temperatures.
- (d) Adhesive tapes or transfer adhesives shall be installed:
 - (i) Between the existing window frame and Fragment Retention Film; and
 - (ii) Between the Fragment Retention Film and the mechanical anchoring system.
- (e) Attachment finishing cap shall have pockets or channels to receive a fixed resilient elastomer gasket between glass and cap to provide sharp appearance and a moisture barrier.
- (f) Attachment finishing cap shall be snapped over attachment baseplate to show only a sharp, uninterrupted exterior. No exposed fasteners, bolts, or adhesive tapes incorporated into the film attachment assembly will be permitted in the finished attachment system except in those facility areas where appearance is not a factor.
- (g) All exposed finishing cap surfaces shall be free of scratches and other serious blemishes.
- (h) All finishing caps shall be:
 - (i) Anodized clear aluminium to match existing frames where required;
 - (ii) Anodized dark bronze to match existing frames where required;
 - (iii) Anodized black to match existing frames where required; and
 - (iv) Coated to match/near match existing framing systems where required.
- (i) Aluminium extrusion finishing cap shall be given a caustic etch followed by an anodic oxide or Kynar paint treatment to obtain:
 - (i) An Architectural Class II clear anodic coating conforming to Aluminium Association Standards;
 - (ii) An Architectural Class I anodic coating with integral/deposited dark bronze or black color conforming to Aluminium Association Standards; and
 - (iii) A factory-applied seventy percent (70%) flouropolymer (Kynar) resin coating or equal conforming to AAMA 2605:
 - 1) Thickness 1.2 mil;
 - 2) Minimum F hardness; and
 - 3) Minimum abrasion coefficient value of 40 per ASTM 968 falling sand test method.

G8.3.64 Fragment Retention Laminate for Window and Glass Panel

- (a) Laminate shall comply with the following criteria:

Glass Breakage Safety	EN 12 600
Tensile Strength MD/TD	224 N/mm ² / 207 N/mm ²
Yield Strength MD / TD	105 N/mm ² / 108 N/mm ²
Elongation at Break MD/TD	100% / 139%
Elongation at Yield MD/TD	7% / 6%
b Value	0.97
g Value	0.83
U-Value (EN 673 W/m ² .K)	5.69
Visible Light Transmission	73.2%
Visible Light Reflectance	19.6%
Ultraviolet Light Transmission	<1%
U-Value Summer	0.88
Shading Coefficient	0.69
Infrared Rejection	51.0%
Solar Energy Transmission	57.05%
Solar Energy Reflectance	31.1%
Solar Energy Absorption	11.85%
Total Solar Energy Rejected	40.0%

G8.3.65 Glass Strengthening Treatment for Window and Glass Panel

- (a) Glass strengthening shall comply with the following criteria:
- (i) Strength Enhancement to Glass by 42%
 - (ii) Impact Energy Ratio = 1.42;
 - (iii) Strength Ratio = 1.40; and
 - (iv) Maximum Strength = 69,800 N (Newton)

END OF SECTION